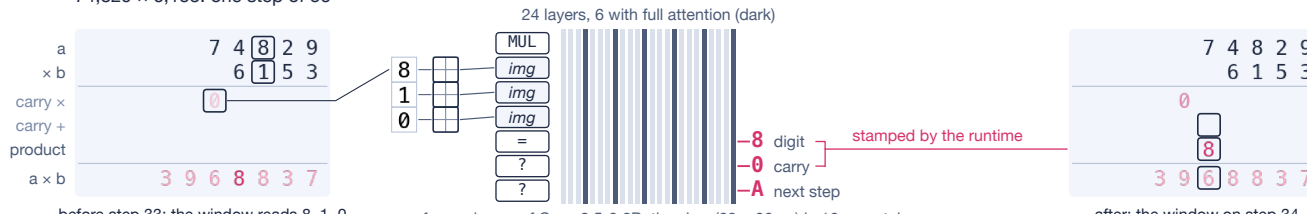
a



before step 33: the window reads 8, 1, 0

one forward pass of Qwen3.5-0.8B: the view (32 × 96 px) in 16 px patches, merged 2 × 2 into three image tokens; the answers are read at = and the two slots

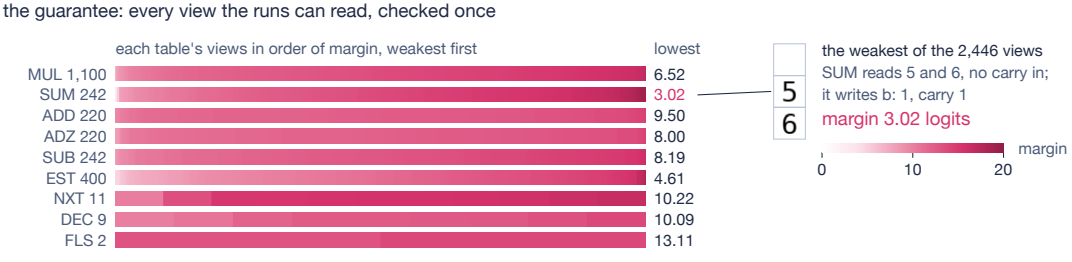
after: the window on step 34



the whole product: 56 steps, one forward pass each, four passes of the multiplier's digits; M multiply A add Z add the last carry F flush N next digit

Between steps the model keeps nothing: the sheet is its only memory.

c



exact on all 2,446 views ⇒ exact on every run of every length (Theorem 1)

2,446 of 2,446 pass on the M4 (WebGPU, fp16); every new GPU checks the weakest tenth again before it runs

d

